Noctua: Unsupervised Network Intrusion Detection under Training Contamination, Temporal Drift and Dataset Shift

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Abstract

Unsupervised detectors are attractive for intrusion detection because attack labels are scarce and attacks evolve, yet they are routinely evaluated on a clean training window, stationary traffic and a single network. We benchmark 15 unsupervised detectors—tree, kernel, density, distance, reconstruction, deep one-class and self-supervised families, including a β-VAE, a RealNVP flow, a neural-transformation model and a masked tabular transformer—on NSL-KDD, UNSW-NB15 and CIC-IDS2017 under a protocol that removes those assumptions one at a time. Every detector is fit without labels and calibrated with split-conformal p-values, which turn any anomaly score into a false-positive guarantee that we then test to breaking. Three findings stand out. First, a few percent of hidden attacks in the training window degrades reconstruction and distance detectors by 11.4 ROC-AUC points on average at $\rho = 5\%$ (up to 36 for kNN distance). Second, *Consensus-Trimmed Refit* (CTR), a label-free screen by three independent detectors, recovers +5.4 points on average at $\rho = 5\%$ and improves the plain detector in 21/21 dataset–detector pairs. Third, conformal calibration is only approximately kept even in-dataset—single detectors realize 3.0–4.9% false positives at a nominal 1%, while a diverse five-detector merge stays within budget (0.2–1.7%) at 41–67% recall—and it breaks under drift and shift: the benign false-positive rate of a Monday-calibrated 1% threshold is 1.6% on the best day of CIC-IDS2017 (Wednesday) and 7.6% on the worst (Thursday), averaged over detectors, and a label-free population-stability index of the benign scores tracks this failure (Spearman $\rho = 0.52$). Cross-dataset transfer between UNSW-NB15 and CIC-IDS2017 is poor (mean AUC 0.695 and 0.446). We release the protocol, per-seed results, an interactive web laboratory, and code.

1 Introduction

Network intrusion detection is a one-class problem in disguise: normal behaviour is abundant, attacks are rare, labelled and non-stationary [2, 25]. Unsupervised and semi-supervised detectors therefore dominate the research literature, but they are usually evaluated under conditions that a deployment cannot guarantee: a curated attack-free training window, test traffic drawn from the training distribution, and a single dataset. When those conditions fail, headline AUCs say little about what an operator experiences, namely the *false-alarm rate* at a threshold chosen in advance.

We study three questions. **(Q1) Contamination:** how much does a small fraction of undetected attacks in the training window hurt each detector family, and can it be repaired without labels? **(Q2) Drift:** how quickly does a threshold calibrated on one day stop meeting its false-positive budget, and is the breakdown visible without labels? **(Q3) Shift:** what survives when a detector trained on one network is deployed on another?

Contributions. (i) A reproducible protocol—label-free fitting, a held-out clean calibration slice, five experiments, bootstrap intervals and rank-based tests—over three public datasets that require no registration. (ii) A conformal layer that gives every detector a finite-sample false-positive bound, a valid merge of heterogeneous detectors, and an empirical account of where the bound breaks. (iii) Consensus-Trimmed Refit (CTR), a simple label-free defence against training contamination with ablations over trim level and screener committee. (iv) A label-free drift alarm validated against calibration breakdown on real timestamps. (v) Exact reconstruction attributions with a deletion-faithfulness test. (vi) An interactive laboratory, per-seed result tables and code.

2 Related work

Anomaly detection benchmarks. Surveys [5, 18] and ADBench [11] compare detectors on tabular data, usually under clean training and i.i.d. splits. Security-specific critiques [2, 25] stress that evaluation must respect time and deployment conditions; TESSERACT [19] does so for malware classification. We apply that discipline to unsupervised network detectors. Datasets used are NSL-KDD [26], UNSW-NB15 [17] and CIC-IDS2017 [24], whose labelling and flow-construction issues are documented [8].

Detectors. We include Isolation Forest [16], one-class SVM [23], LOF [4], ECOD [15], COPOD [14], HBOS [9] through PyOD [29], a Deep SVDD [22], an autoencoder, a β-VAE [12, 13], a RealNVP flow [7], NeuTraL AD [20] and a masked feature-reconstruction transformer in the spirit of FT-Transformer [10].

Robustness and conformal methods. Robust autoencoders [30] handle corrupted training data inside one model; CTR instead screens the data with a committee before any model is fit. Conformal prediction [1, 28] yields distribution-free guarantees; conformal p-values for outlier testing are analysed by Bates et al. [3], and p-values can be merged by averaging [27]. Drift detection on model outputs is studied by Rabanser et al. [21].

3 Setting and protocol

Data. NSL-KDD (125,973 / 22,544 flows, 41 features), UNSW-NB15 (official 175,341 / 82,332 split, 34 features) and CIC-IDS2017 (2.83M flows, 78 features, timestamps), the latter subsampled class-aware (rare classes kept whole) and split temporally: Monday–Tuesday for training (Monday is attack-free), Wednesday–Friday for testing, so most test attack families are novel. For cross-dataset experiments UNSW-NB15 and CIC-IDS2017 are mapped to 18 physically aligned features. Identifiers (IPs, ports, flow id) are never features. All downloads are pinned by SHA-256 from public mirrors.

Contract and calibration. A detector fits on unlabeled rows and returns a score, higher meaning more anomalous. Labels only score results. In every experiment 15% of the training split's clean normal flows (at most 20k) are held out as a calibration set $\mathcal{C}$; conformal p-values are $p(s) = (1 + |\{c \in \mathcal{C} : c \geq s\}|)/(|\mathcal{C}| + 1)$, so that if calibration and test normals are exchangeable, $\Pr(p \leq \alpha) \leq \alpha$ for any detector [28]. The average $\min(1, 2\bar{p})$ of several such p-values is again valid [27]; for ranking we use the mean of $-\log_{10} p_k$. No hyperparameters are tuned on labels: every detector uses one fixed configuration across datasets.

Experiments. **E1** main benchmark: all detectors, 3 datasets, *clean* (normal-only) and *natural* (training split as released) regimes, 5 seeds. **E2** contamination: attacks injected into a 30k-row normal base at $\rho \in \{0, 1, 2, 5, 10, 20\}\%$. **E3** drift: fit and calibrate on Monday, evaluate Tuesday–Friday. **E4** shift: train on one dataset's harmonized clean normals, detect on the other, with source or per-domain standardization. Statistics follow Demšar [6]: Friedman tests, Nemenyi critical differences and Holm-corrected Wilcoxon tests over (dataset, regime, seed) blocks; because dataset splits are fixed, seeds are replicates over models and the tests are optimistic, which we say wherever they are used.

4 Consensus-Trimmed Refit

Attacks hidden in the training window are learned as normal by density, reconstruction and distance models. CTR removes them *before* fitting: three structurally different cheap screeners (Isolation Forest, ECOD, PCA reconstruction) are fit on the training rows and score them; each score is converted to a percentile rank, the ranks are averaged, and the top τ fraction by consensus rank is dropped; the base detector is then fit on the remainder. The knob τ is an assumed upper bound on contamination. Agreement across independent inductive biases is a more specific contamination signal than any one detector's tail, which we test by ablation.

5 Results

E1: main benchmark. Table 1 gives ROC-AUC; Fig. 1 the critical-difference diagram (Friedman $\chi^2 = 85.6$ over 30 blocks, CD = 3.92); the best-ranked single detector is Autoencoder. The best clean-regime single-detector AUC is 0.966. Calibration is the more telling comparison (Fig. 4). At a nominal $\alpha = 1\%$ the realized false-positive rate of single detectors in the clean regime is 3.0% on NSL-KDD, 2.1% on UNSW-NB15 and 4.9% on CIC-IDS2017: even in-dataset, test traffic is shifted relative to the calibration slice (novel

	NSL-KDD-C	UNSW-NB-C	CIC-IDS-C	NSL-KDD-N	UNSW-NB-N	CIC-IDS-N
Autoencoder	**0.966**	0.913	0.864	0.804	0.315	0.863
Gaussian mixture	0.959	0.906	0.873	0.900	0.259	0.877
kNN distance	0.962	0.916	0.844	0.646	0.439	0.843
NeuTraL AD	0.951	**0.931**	0.831	0.764	0.439	0.844
β-VAE	0.962	0.863	**0.880**	0.809	0.251	**0.877**
Deep SVDD	0.957	0.889	0.855	0.730	0.402	0.849
RealNVP flow	0.933	0.903	0.853	0.823	0.285	0.854
Masked Tab-Transformer	0.930	0.859	0.861	0.726	0.369	0.856
PCA reconstruction	0.955	0.798	0.861	0.815	0.417	0.861
Local Outlier Factor	0.865	0.825	0.870	0.642	**0.529**	0.867
Isolation Forest	0.949	0.781	0.811	0.870	0.282	0.808
HBOS	0.945	0.674	0.724	0.853	0.293	0.722
ECOD	0.952	0.588	0.731	**0.903**	0.329	0.734
One-Class SVM	0.886	0.590	0.789	0.840	0.439	0.794
COPOD	0.949	0.550	0.719	0.897	0.290	0.722
Ensemble (all)	0.961	0.897	0.877	0.872	0.328	0.878
Ensemble (diverse 5)	0.960	0.920	0.852	0.821	0.327	0.857

Table 1: ROC-AUC (mean over seeds). Columns: dataset and regime (C = clean, N = natural training). Bold: best single detector per column; ensembles listed last.

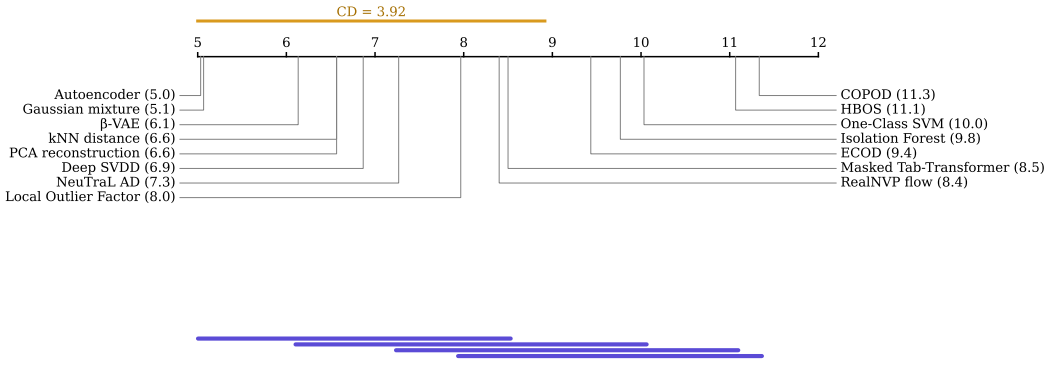


Figure 1: Critical-difference diagram (ROC-AUC). Methods joined by a bar are not significantly different (Nemenyi, α=0.05).

services, later days). Merging detectors with averaged conformal p-values is valid but its power depends on its members: the diverse five-detector merge (Isolation Forest, kNN, AE, flow, NeuTraL) stays within budget (0.2–1.7%) with 41–67% recall at $\alpha = 5\%$, whereas merging all fifteen—including weak members such as ECOD and COPOD on UNSW-NB15—is so conservative that recall at $\alpha = 5\%$ falls to 1.4% there.

E2: contamination. Fig. 2 and Table 2 show the effect of hidden attacks. At $\rho = 5\%$ the mean AUC drop is 11.4 points. CTR with a 10% trim changes the AUC by +5.4 points on average (improving 21/21 dataset–detector pairs); the gain is +4.8 at $\rho = 2\%$ (21/21), +2.2 at 10% (16/21) and +1.4 at 20% (13/21), and the price on clean data ($\rho = 0$) is -2.5 points because rare-but-legitimate flows are trimmed. The ablation (Table 3) compares trim levels, single screeners and an oracle trim equal to the true ρ.

E3: drift. Fig. 3 (left) shows the realized benign false-positive rate by day for a Monday-calibrated 1% threshold. Drift is *not* monotone in time: averaged over single detectors the rate is 1.6% on Wednesday and 7.6% on Thursday, because each day adds its own benign services. The population-stability index of benign scores, which needs no labels, correlates with the realized rate (Spearman $\rho = 0.52$, Fig. 3 right). A PSI$\geq$ 0.10 alarm flags the breakdown (FPR $\geq 3\times$ nominal) with recall 58% and precision 76%; the stricter PSI$\geq$ 0.25 threshold has precision 79% but recall 13%, so the monitor is a conservative early warning rather than a complete one.

E4: dataset shift. Fig. 5: in-dataset mean AUC is 0.667 (UNSW) and 0.757 (CIC); transferred it is 0.695 (UNSW→CIC) and 0.446 (CIC→UNSW), with realized FPR at nominal 1% of 56% and 28%. Per-domain

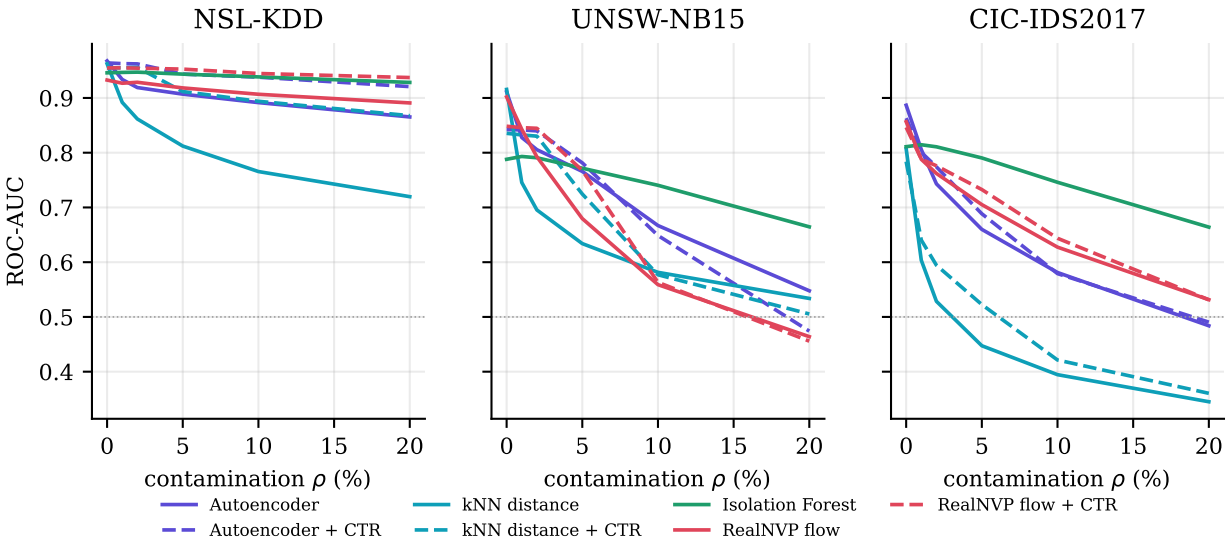


Figure 2: ROC-AUC vs. training contamination. Solid: plain detector; dashed: behind CTR (trim 10%).

Dataset	Detector	ρ=0	ρ=5%	ρ=5% + CTR	Δ
NSL-KDD	Autoencoder	0.968	0.907	0.944	+0.037
NSL-KDD	kNN distance	0.962	0.812	0.912	+0.100
NSL-KDD	RealNVP flow	0.933	0.918	0.953	+0.034
NSL-KDD	Gaussian mixture	0.964	0.924	0.953	+0.028
UNSW-NB15	Autoencoder	0.911	0.766	0.781	+0.016
UNSW-NB15	kNN distance	0.916	0.634	0.725	+0.091
UNSW-NB15	RealNVP flow	0.902	0.680	0.768	+0.088
UNSW-NB15	Gaussian mixture	0.901	0.651	0.719	+0.068
CIC-IDS2017	Autoencoder	0.887	0.660	0.688	+0.029
CIC-IDS2017	kNN distance	0.807	0.447	0.523	+0.075
CIC-IDS2017	RealNVP flow	0.857	0.705	0.733	+0.028
CIC-IDS2017	Gaussian mixture	0.890	0.754	0.805	+0.051

Table 2: Effect of 5% contamination and CTR (mean over seeds).

standardization changes the transferred AUC to 0.610 and 0.550.

Explanations. For the autoencoder, per-feature squared error is an exact additive decomposition of the score. Deleting the top-3 attributed features removes 51%, 15% and 38% of the score on NSL-KDD, UNSW-NB15 and CIC-IDS2017, against 2%, 1% and 3% for random features (Fig. 6); k-means on attribution vectors of flagged flows recovers attack categories with NMI 0.19 / 0.41 / 0.46.

6 Discussion and limitations

Conformal calibration does what it promises when exchangeability holds, and it is precisely the diagnostic that reveals when it does not; this is more useful operationally than another AUC. Limitations (see the repository's `docs/limitations.md`): all datasets are lab-generated and have documented issues; fixed hyperparameters are fair across datasets but not optimal per dataset; seeds do not replicate datasets; CTR needs an assumed contamination bound; the harmonized transfer space is lossy; and we do not evaluate adaptive adversaries. Tabular foundation models such as TabPFN were examined for feasibility and excluded because per-feature conditional prediction is impractical on CPU at this scale.

7 Conclusion

Evaluating unsupervised intrusion detectors on a clean training window with a single threshold-free number hides exactly the failure modes that matter in deployment. A protocol that injects contamination, replays

Purification of the AE	NSL-KDD	UNSW-NB15	CIC-IDS2017
none	0.907	0.766	0.660
CTR, trim 5%	0.947	0.737	0.662
CTR, trim 10% (committee)	0.944	0.781	0.688
CTR, trim 20%	0.960	0.817	0.697
screener: IForest only	0.964	0.807	0.683
screener: ECOD only	0.961	0.827	0.684
screener: PCA only	0.933	0.678	0.646
oracle trim $= \rho$	0.947	0.737	0.662

Table 3: Ablation of CTR on the autoencoder at $\rho = 5\%$ (ROC-AUC).

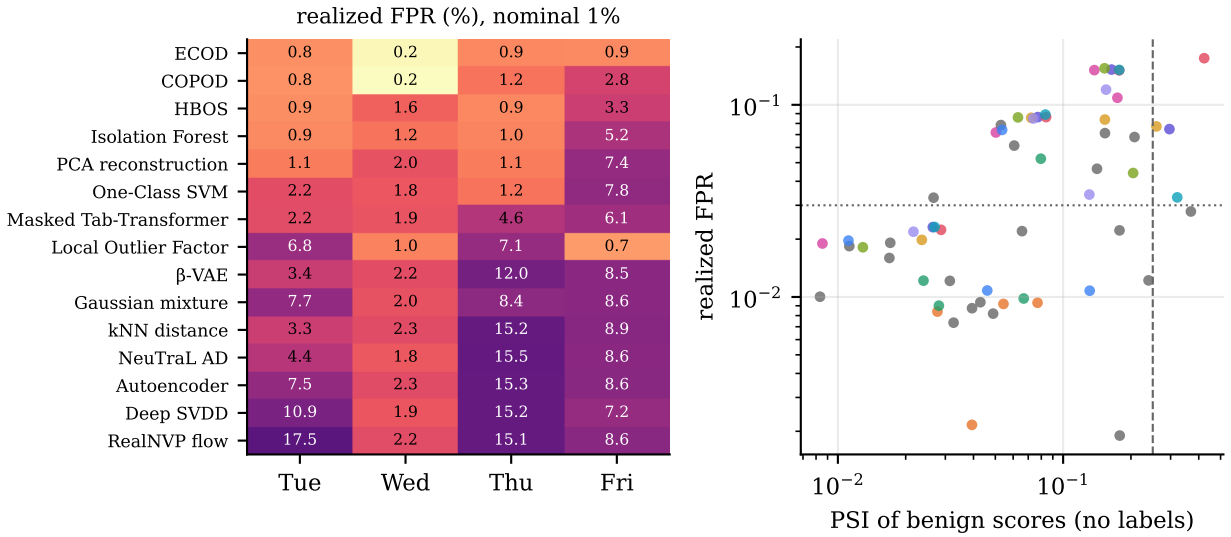


Figure 3: Left: realized FPR (%) at nominal 1% by day. Right: label-free PSI vs. realized FPR (dashed lines: PSI 0.25, FPR 3%).

time and crosses networks, paired with conformal calibration and a label-free drift monitor, makes those failures measurable. Code, per-seed result tables and an interactive laboratory accompany this paper; raw score files are regenerated by one command.

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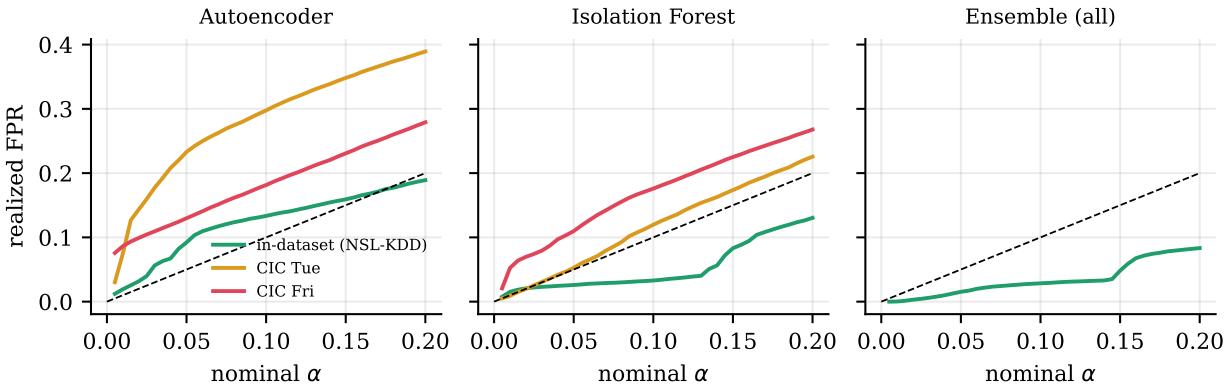


Figure 4: Reliability diagrams: realized vs. nominal false-positive rate. On the diagonal the conformal guarantee holds; above it, it is broken.

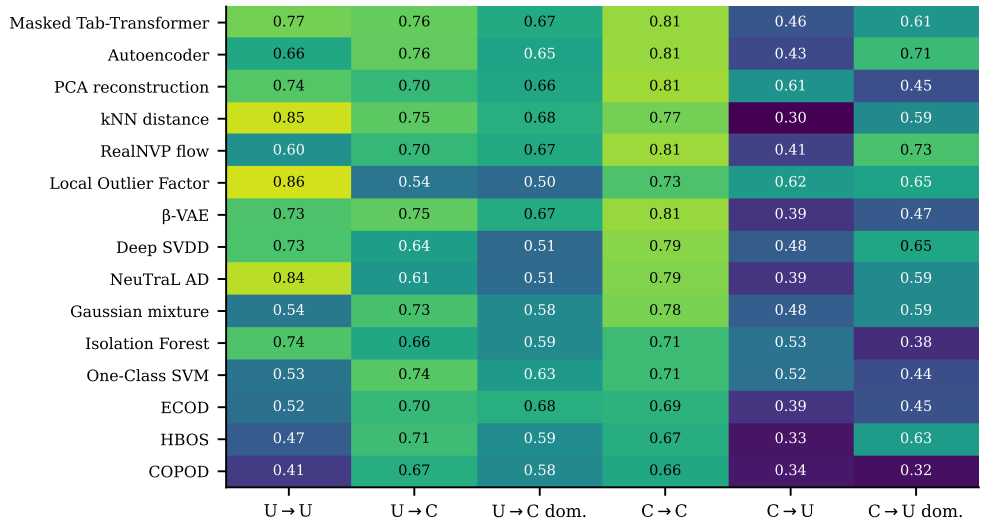


Figure 5: ROC-AUC for in-dataset and cross-dataset evaluation on the 18-feature harmonized space (dom. = per-domain standardization).

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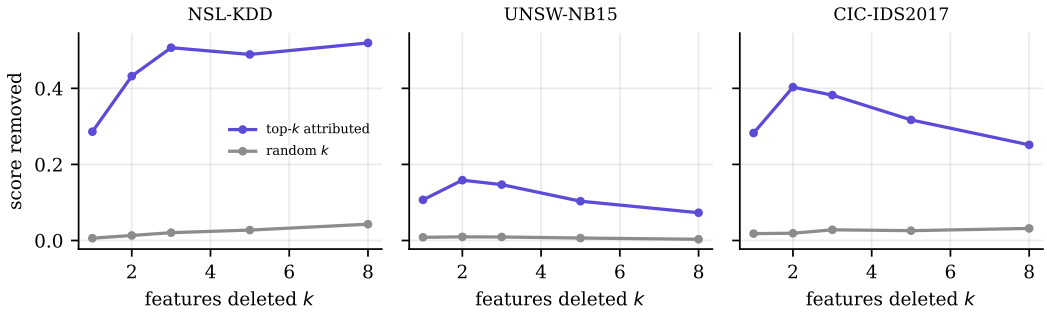


Figure 6: Deletion faithfulness of autoencoder attributions.

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